

### Inverse axiom (3)

$$\begin{array}{ccc} G & \xrightarrow{(\mathrm{id}_G, \nu)} & G \times G \\ \downarrow & & \downarrow \mu \\ * & \xrightarrow{e} & G \end{array}$$

$$\begin{array}{ccc} G & \xrightarrow{(\nu, \mathrm{id}_G)} & G \times G \\ \downarrow & & \downarrow \mu \\ * & \xrightarrow{e} & G \end{array}$$